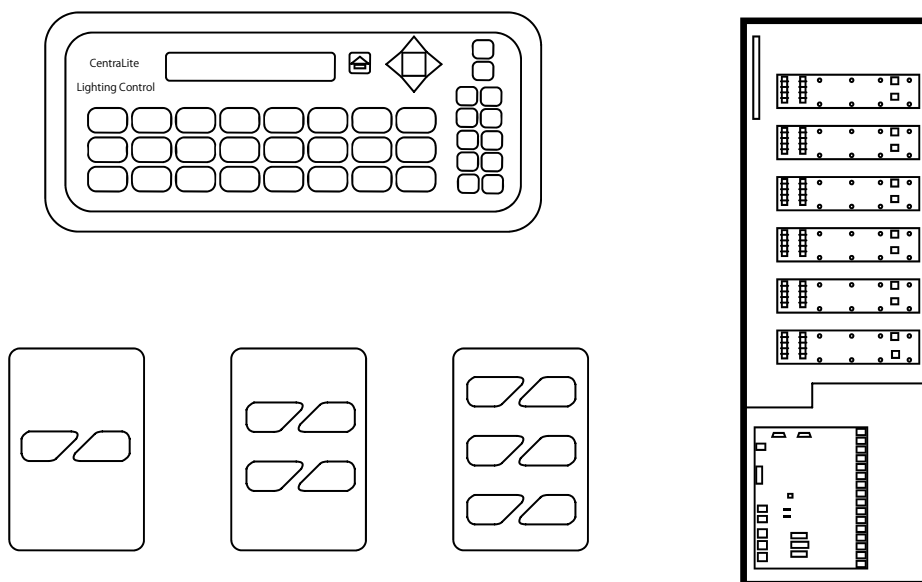




SIMPLE. RELIABLE. AFFORDABLE.

CL24 LiteJet™

UK Installation and Programming Guide



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CL24 LiteJet™ Specifications and Load Ratings

Types of Acceptable Power: Single or Three Phase (Optional) Power @ 50 or 60 Hertz

Relay Panel Max. Load: 72 Amps

Relay Module Max. Load: 16 Amps

Individual Relay Max. Load: 8 Amps

Allowable Types of Dimming: Incandescent, Magnetic Low Voltage, and Suitable Electronic Low Voltage Fixtures

Dimming Method: Normal Phase Control leading edge dimming.

Note: Use copper conductors only.

Max. Ambient Temperature: 40 Degrees Celsius

Min. Ambient Temperature: 0 Degrees Celsius

Relay Panel must be mounted upright.

Do not insulate within 6 inches of Relay Panel.

All components must be mounted in dry conditions. Do not expose to rain, high humidity, or other sources of moisture.

DO NOT EXCEED 72 AMPS TOTAL RELAY PANEL LOAD

DO NOT EXCEED 16 AMPS LOAD ON ANY RELAY MODULE

DO NOT EXCEED 8 AMPS LOAD ON ANY INDIVIDUAL RELAY

The **CL24 LiteJet™** is a centralized lighting control system. The **CL24 LiteJet™** can control electrical devices other than Lights, but since most of what the **CL24** controls are Lights, that's what we'll talk about.

The **CL24 LiteJet™** controls Lights with solid-state **Relays**. The **Relays** are attached to Relay Modules. (**Page 7**) The Relay Modules are enclosed in the top of the Relay Panel. (**Page 6**)

The **Relays** are controlled by the Master Control Panel (**MCP**). The **MCP** is located at the bottom of the Relay Panel. (**Page 6**)

The **MCP** receives control signals from **Buttons**. Buttons are located on **Keypads**. The Keypads are placed on the walls where switches would normally go. Keypads are connected to the MCP with CAT5 cables (*stranded* CAT5e patch cable is recommended to avoid conductor breakage). (**Page 9**)

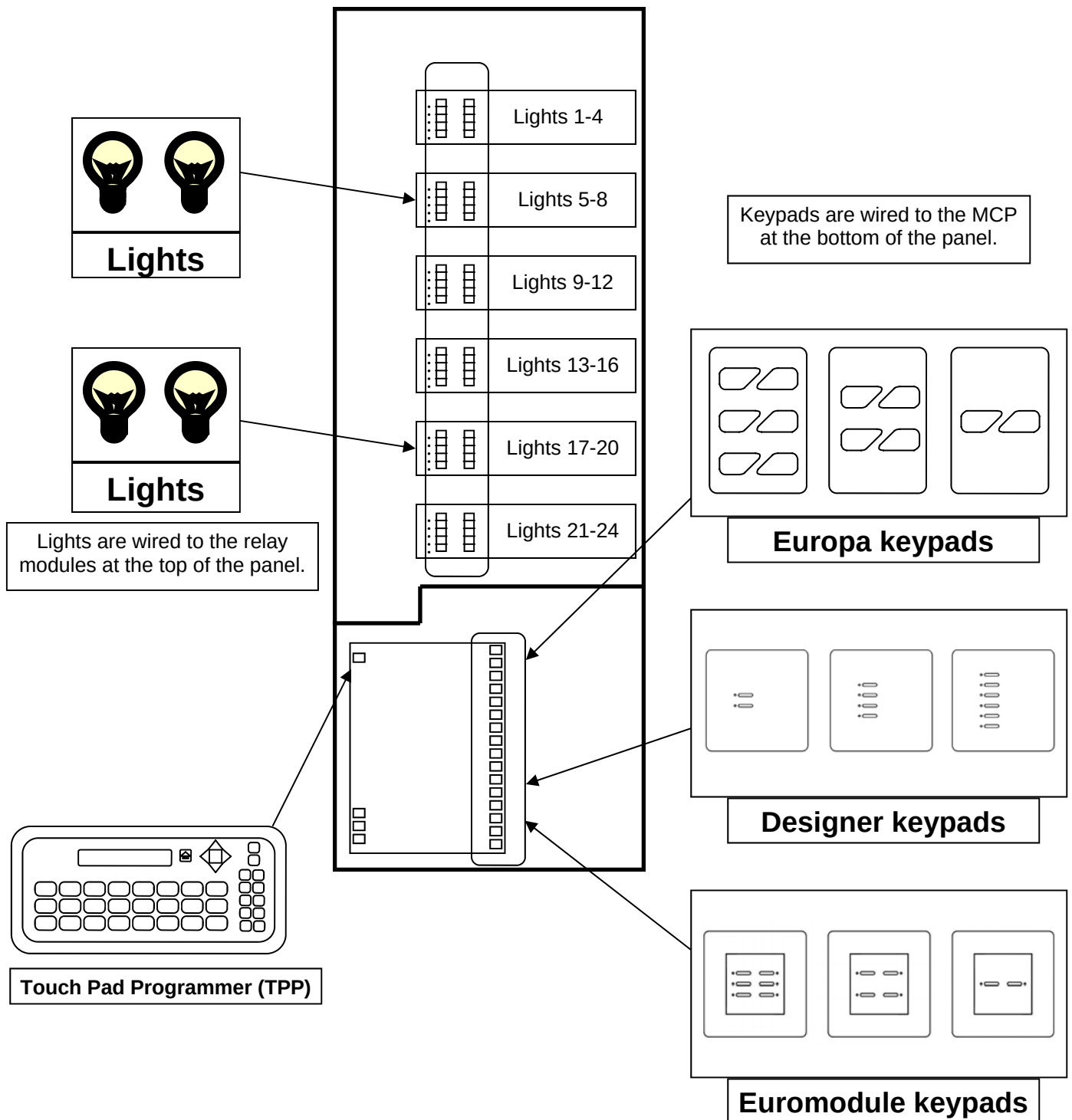
Buttons can control individual **Lights** or groups of **Lights** in **Scenes**. (**Page 18**)

Scenes are groups of Lights which are preset to turn on to various levels of dimming. (**Page 23**)

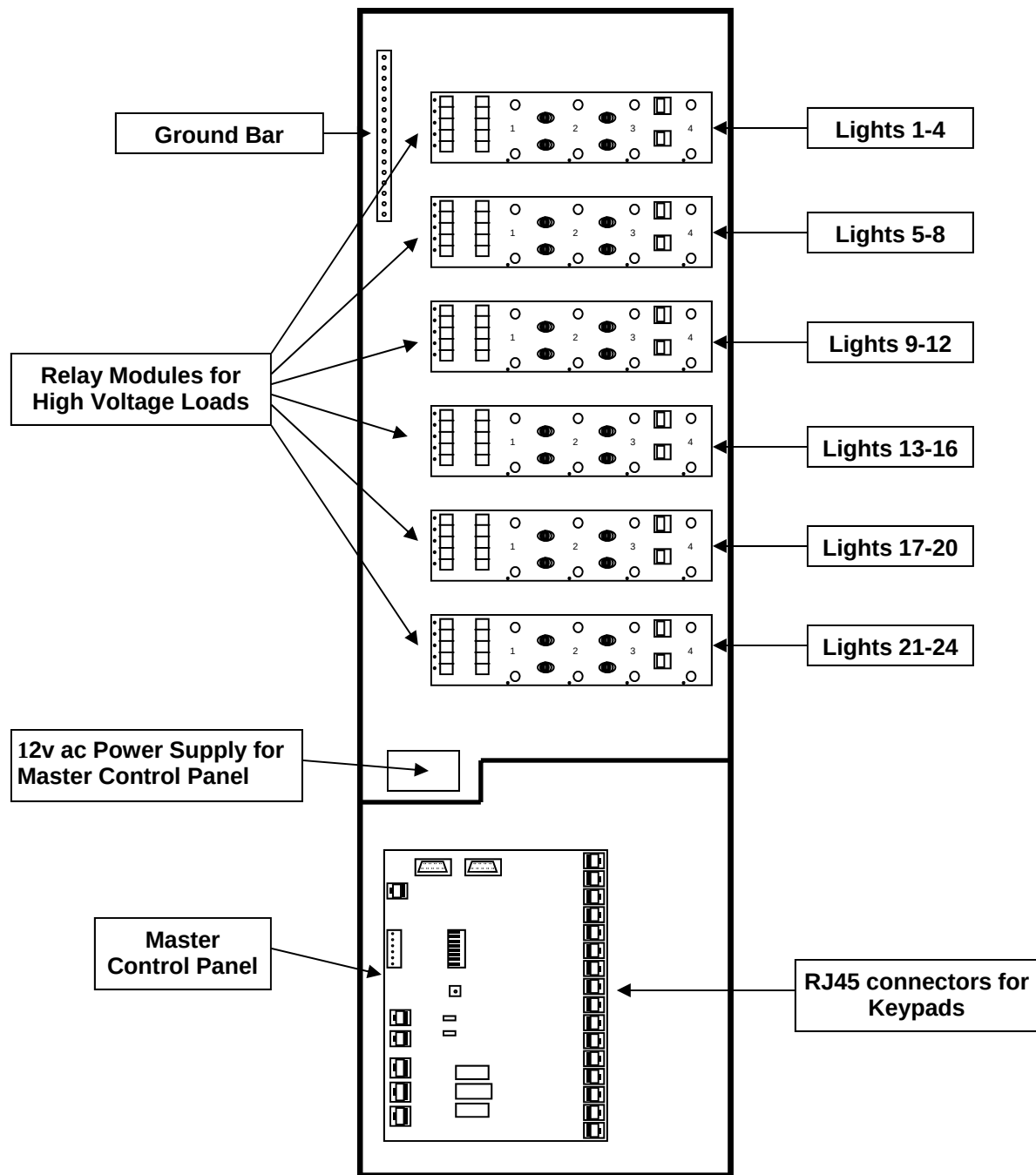
The system is programmed from the Touch Pad Programmer (**TPP**). The **TPP** is placed somewhere in the home. The TCP is connected to the MCP with CAT5 cable. The system can also be programmed from a PC using the **LiteJet™** software. The system can operate without a **TPP**.

The **CL24 LiteJet™** can interface with other control systems and devices through its RS232 and RS485 ports.

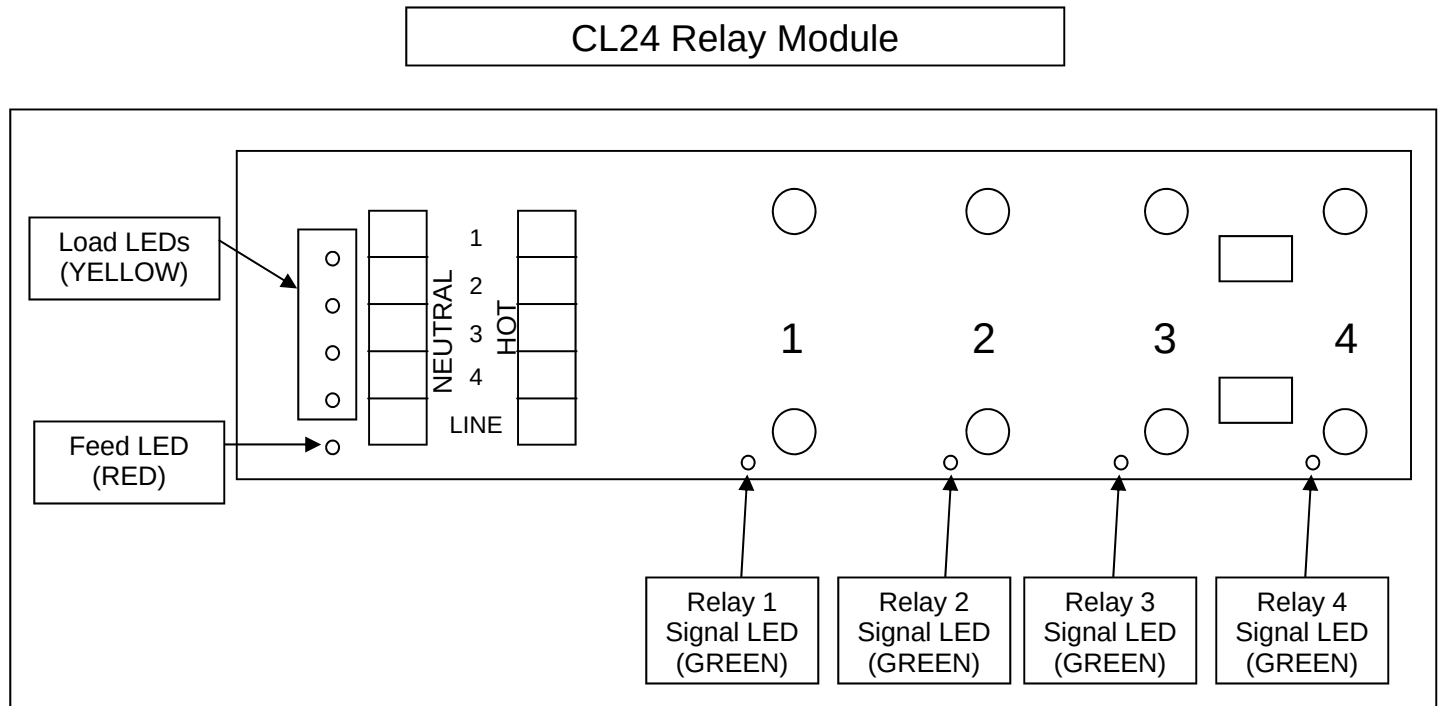
CL24 LiteJet™ System Diagram



CL24 LiteJet™ Relay Panel Layout



CL24 LiteJet™ High Voltage Wiring

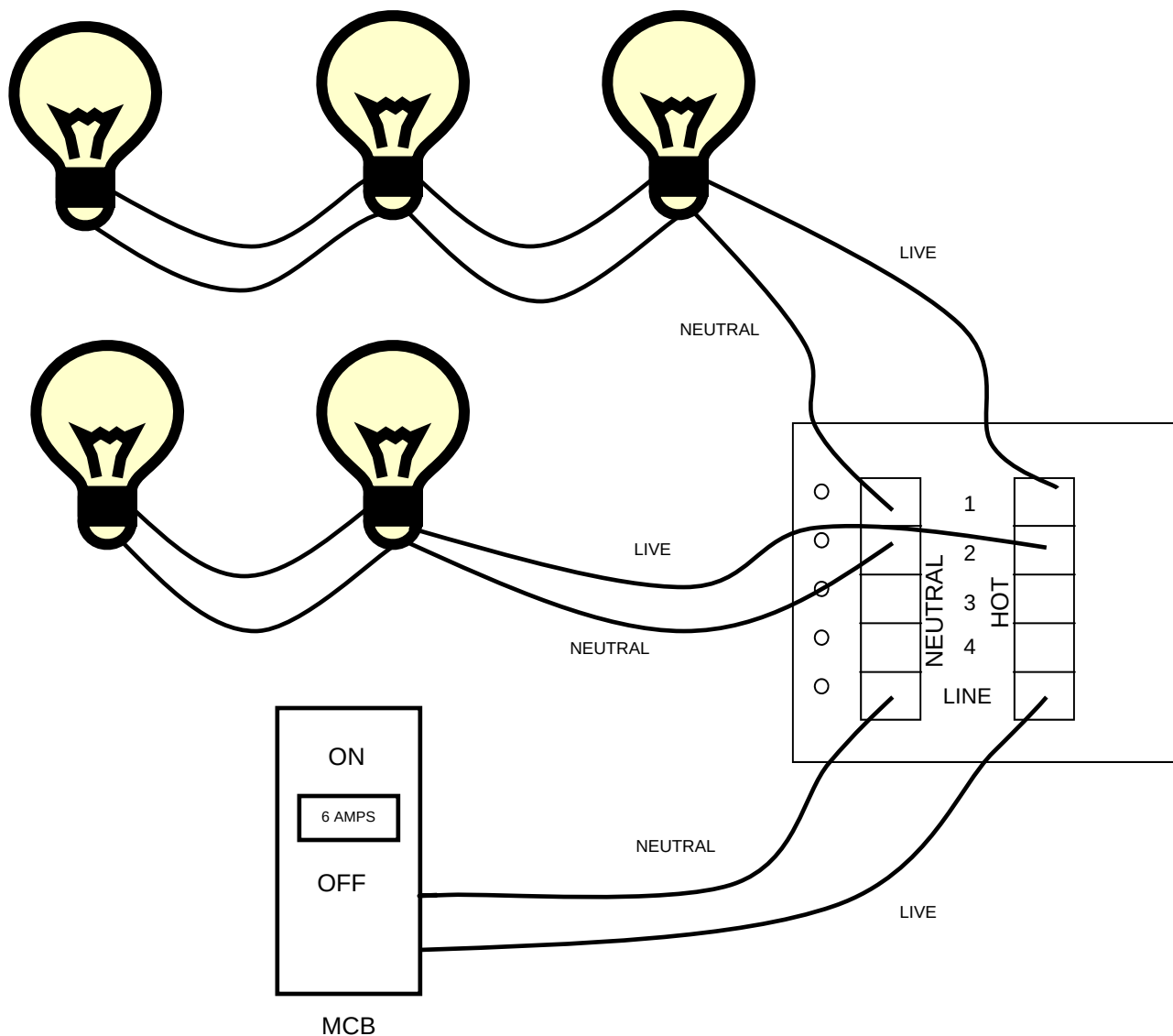


Each CL24 relay module contains 4 solid-state relays. Each module has status LED's which indicate information about the low voltage signal to the relays, the feed power to the module from the breaker, and the high voltage output of each relay. One look at the relay module yields information that usually requires a voltmeter and test light.

If the **RED** Feed **LED** is lit, this means power from the breaker is feeding the module.

If the **GREEN** **LED** for a relay is lit, this means that a 5 Volt signal is being sent to the relay from the MCP.

If the **YELLOW** **LED** is lit, this means the relay is outputting high voltage from the terminal and the light fixture it is connected to should be working.



Each Lighting Switch Leg is terminated on a LOAD terminal. Each LOAD LIVE wire is terminated on it's appropriate LIVE side. Each LOAD NEUTRAL is terminated to it's appropriate NEUTRAL side.

All grounds are terminated from the breakers and the loads on the GROUND BAR.

DO NOT load Relay Panel with more than 72 Amps.

DO NOT control more than 16 AMPS with any one 4 RELAY MODULE.

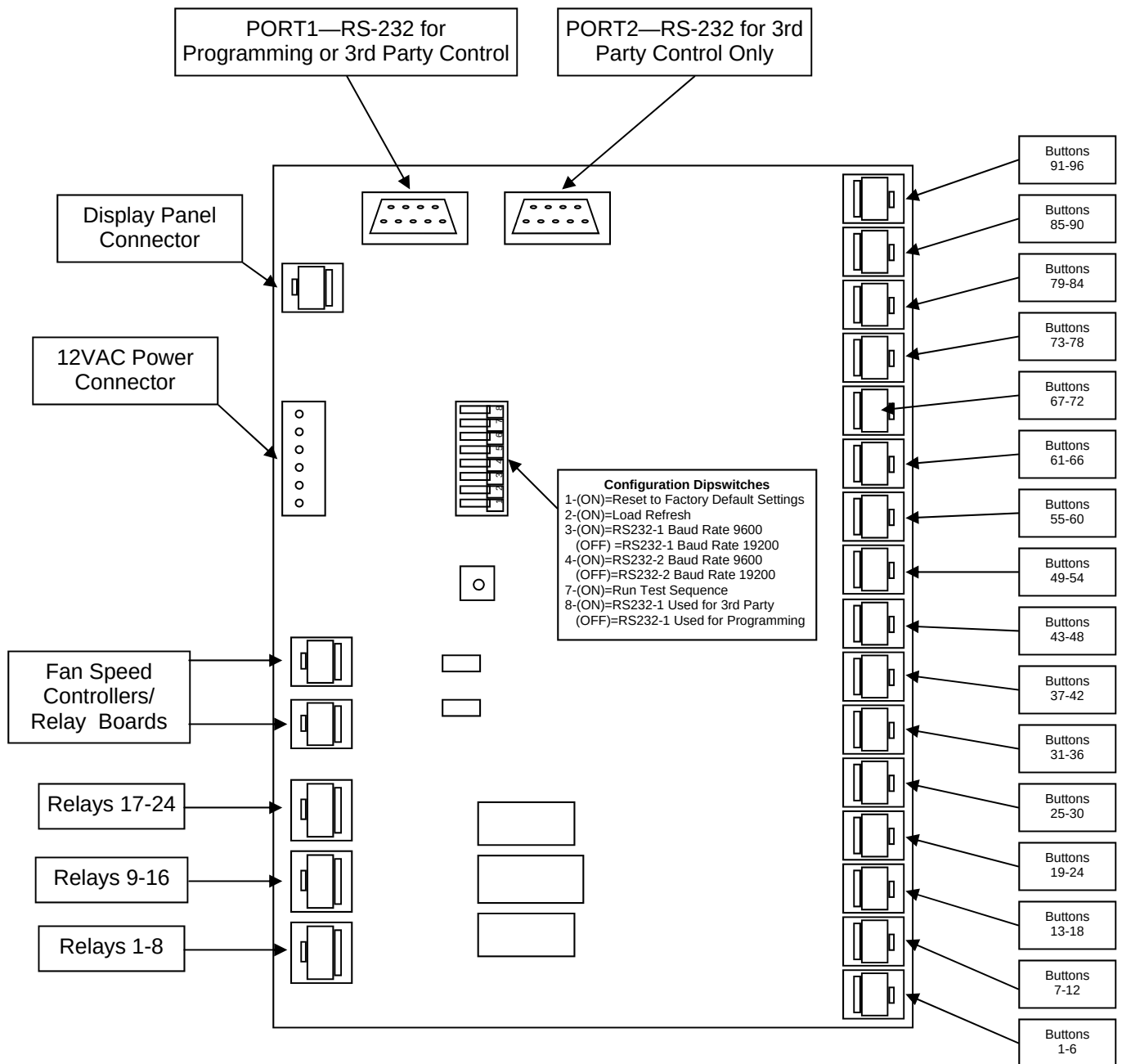
DO NOT control more than an 8 AMP load from any one RELAY.

(Exception: Each of the lower 2 Relay Modules contains a Relay marked with a Purple dot. These relays can be run at a maximum of 12 Amps.)

CL24 LiteJet™ Master Control Panel Description

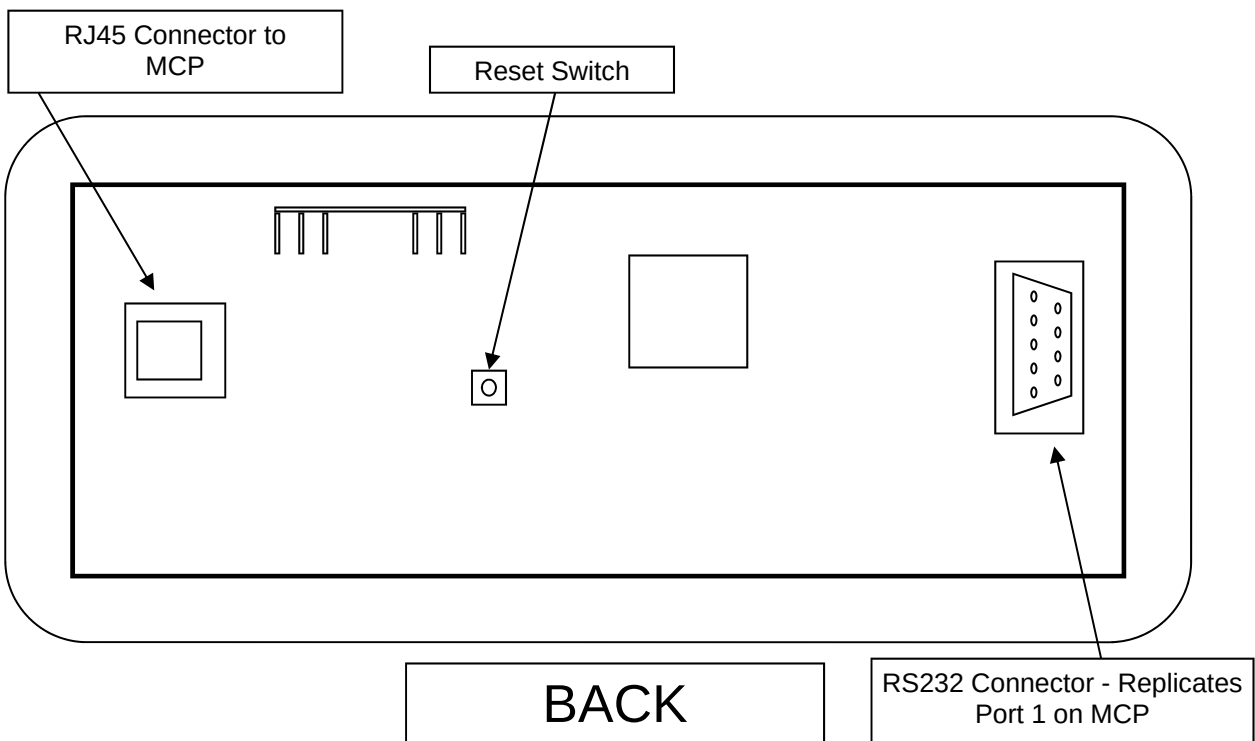
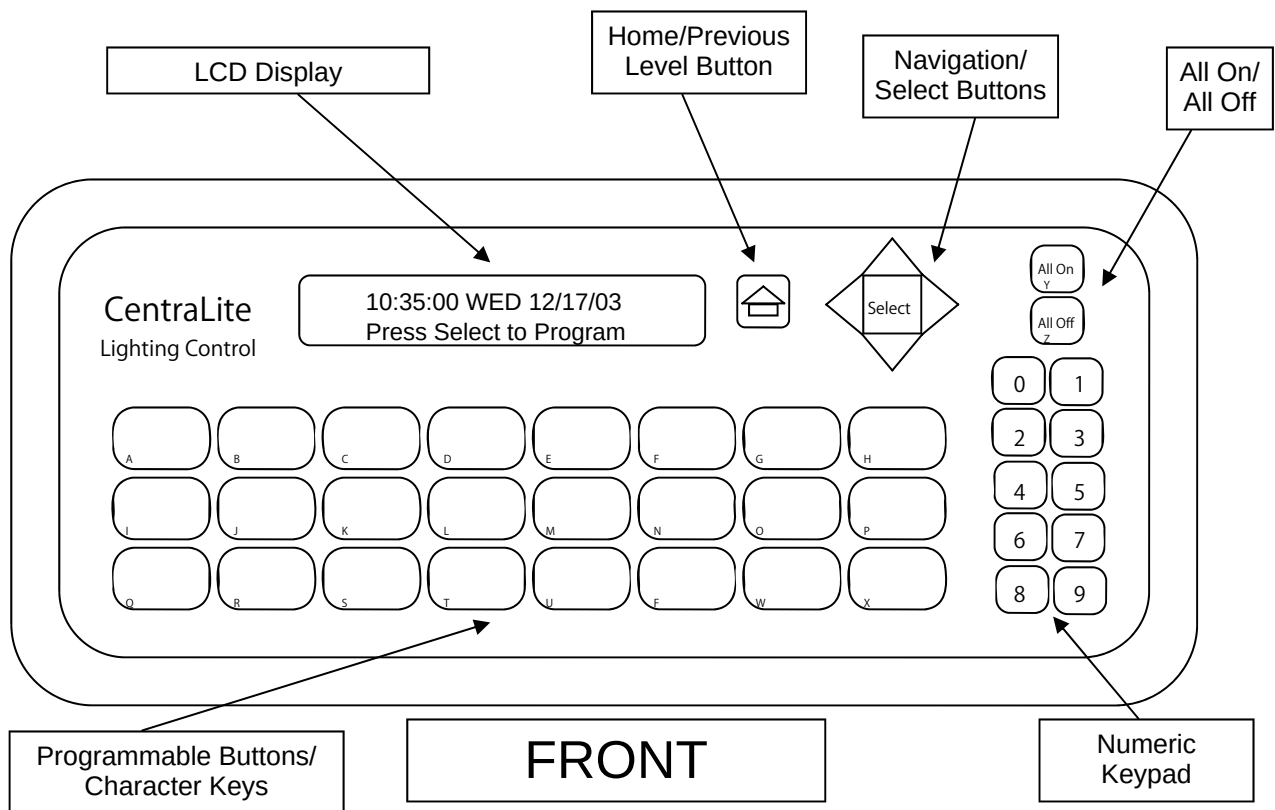
The CL24 Master Control Panel (MCP) is shown below. The MCP consists of :

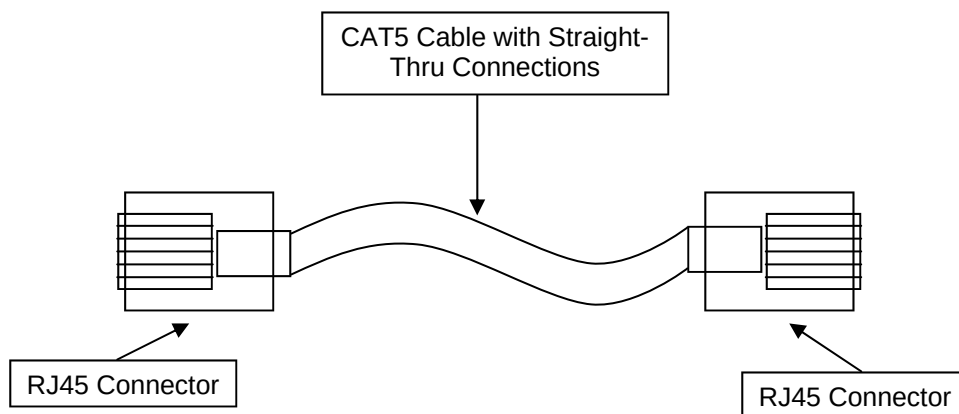
- 16 RJ45s(8-pin) connectors for wall keypads
- 2 RS232 ports for Programming and Third Party Control
- 1 RJ45(8 Pin) for the Display Panel
- 3 RJ45s(10 pin) for connections to the relay modules
- 2 RJ45s(8 Pin) for optional Fan Speed Controllers and Relay Boards



CL24 LiteJet™ TouchPad Programmer

The CL24 Lighting control system can be programmed either from the Touch Pad Programmer (TPP) or a PC connected via RS-232. This guide focuses on programming from the TPP.





Almost all connections between components of the CL24 are made with CAT5 cable with 8-pin RJ45 jacks and plugs. These include connections between the keypads(*) and MCP, Touch Pad Programmer and MCP, and Fan Speed Controllers/Relay Boards and MCP.

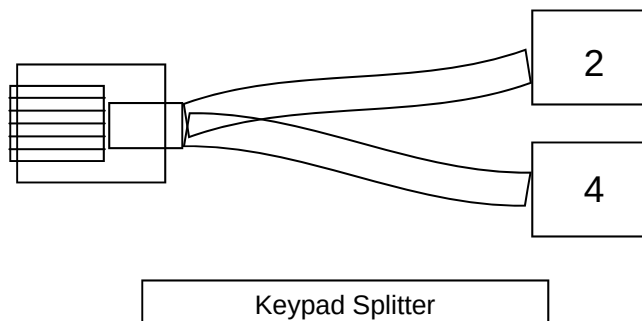
These CAT5 cables are all made with a Straight through wiring termination. (Pin 1 goes to Pin 1, Pin2 goes to Pin 2, etc.) -Use the TIA 568A wiring standard colour codes. See page 41 for details.

(*) UK "Designer" and "Euromodule" keypads do not have RJ45 connectors, they utilise screw down connections - Please refer the connection details shown on page 41. **Stranded** CAT5e patch cable is strongly recommended to avoid conductor breakage.

It is a good idea to use a cable tester to check proper pin-out and continuity between cable ends before connecting devices. This is especially important with the Touch Pad Programmer as a incorrectly wired cable can damage the MCP when connected to the TPP.

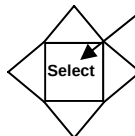
Each keypad connector on the MCP is an RJ45. There are 16 connectors on the MCP. This means that ,normally, up to 16 keypads can be connected to the MCP. However, this is sometimes not enough connectors.

For example, if you have 18 keypads in a single CL24 system, you can still make this work. You would use 2 keypad splitters to connect all 18 keypads. A keypad splitter makes 1 Keypad jack into 2. Each keypad jack on the MCP can support up to 6 buttons. A splitter allows you to connect a 2 button keypad on one split jack and a 2 or 4 button keypad on the other.



CL24 LiteJet™ Programming Overview

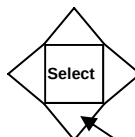
10:35:00AM Wed 12/17/03
Press Select to Program



When the CL24 is in normal operation the screen above will be visible on the TouchPad Programmer. This screen indicates the current time and date held by the CL24 clock. The screen also displays: **“Press Select to Program.”**

To get to the initial programming screen, just press **SELECT**.

1.Lights 2.Buttons
3.Scenes 4.Clock ▼



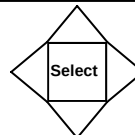
This is the initial programming screen. You can choose to program **Lights**, **Buttons**, **Scenes**, or the **Clock**. You can also see more options by pressing the “down” arrow key(▼) .

The only way to exit the programming mode is to hit the (HOME) key.

At any point in programming, you can go back one level by pressing .

If you ever lose your place or want to exit programming, juts press until you get back to the time and date screen.

LT 01 ▼:Next ▲:Prev
Dimmer? ◀:Yes ▶:No



Any time you see a ◀, ▶, ▼, or ▲. This means that you have the choice of pressing the LEFT, RIGHT, DOWN, or UP keys to navigate or make a choice.

Sometimes, you will see this: (0,1,,9). This means you can enter numbers with the numeric keypad.

Pressing the Select key will typically advance to the next step in programming. This is not the case when entering text. The SELECT acts as a SPACE key when entering names of Lights, Switches, or Scenes.

You will occasionally see a blinking rectangular cursor over the first letter of a choice which represents the current setting. This cursor will also show you which character is being set when entering text.

CL24 LiteJet™ Programming Lights

The CL24 system is used primarily to control lighting. It can also control fans, mechanical relays, and almost any other electrical devices. The CL24 uses dimmable solid state relays to control these devices. The solid-state relays are mounted on relay modules inside the CL24 Relay panel. These modules should only be installed or serviced by a qualified electrician.

They contain high voltage electricity.

(Page 8) features a diagram of a single relay control module. The module contains 4 solid-state dimming relays. At the left end of each module are two wire terminals. One is for LIVE (Red/Brown) wires. The other is for NEUTRAL (Black/Blue) wires. The lighting zones are labeled 1 through 4. At the bottom of each connector is a LINE terminal. The LIVE and NEUTRAL feeds from the breaker are connected to the LINE terminals on each connector.

The installing electrician must ensure that all High Voltage wiring complies in ALL respects with the requirements of BS7671 - requirements for Electrical installations current edition

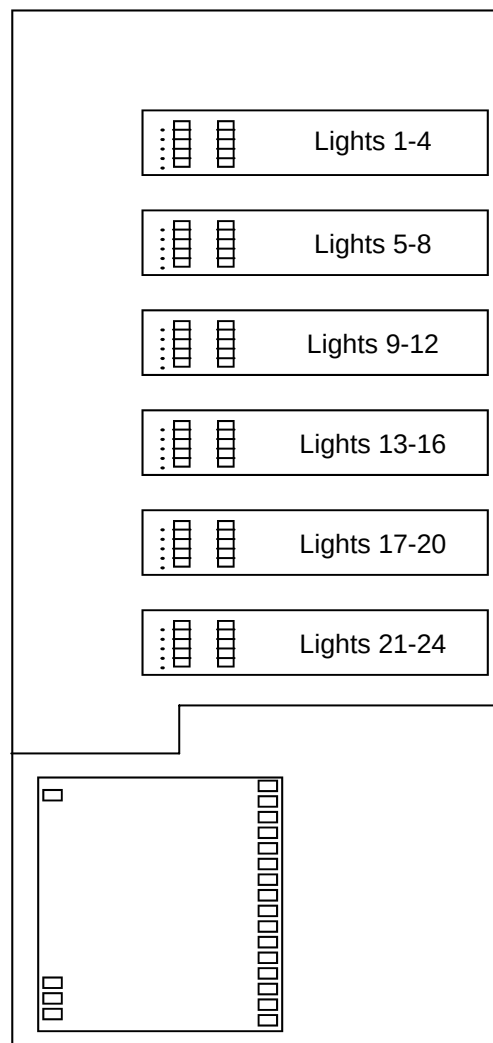
DO NOT apply high voltage insulation tests to this equipment.

Disconnect ALL LIVE and NEUTRAL wires before any insulation tests are applied.

A Light controlled by the CL24 is addressed according to where it is connected.
The following diagram shows how Lights are addressed. Each 4 zone Relay module is marked 1 through 4.

In programming, the lights are addressed from 1 through 24.

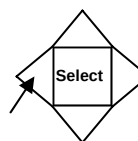
It is very important to verify which light numbers correspond to which light fixtures and label them appropriately.
This is the first step in programming.



CL24 LiteJet™ Programming Lights

STEP 1

LT ▲ 01▼ "Light #1"
◀:Identify ▶:Edit



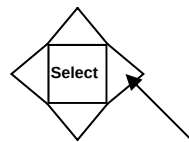
We will begin with programming **Lights**. If this is the initial programming of the system, then the **Lights** will not have names. It is important to give every **Light** a name for reference later.

If you already know what Light #1 is, then you can go straight to **EDIT** to change the **Light** settings.

If you do not know what Light #1 is controlling you can press **Select** to blink the light on and off. Now you can walk around and find which Light is blinking. It is always a good idea to verify **Lights** even if your using a list for programming.

STEP 2

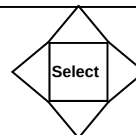
LT ▲ 01▼ "Light #1"
◀:Light Blinking ▶:Edit



As an example, we found that the Garage lights were blinking. We press ▶ to Edit this Light's settings.

STEP 3

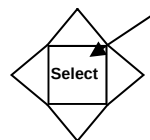
LT 01 SEL=Space ▼:Next
Name: "Light #01"



The first setting to change is the name of the Light. This is very important because we will refer to this Light in other areas of programming. It is much easier to refer to the Lights when they have a name attached.

STEP 4

LT 01 SEL=Space ▼:Next
Name: "Garage"

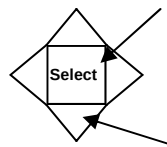


Now we type in the name for this Light: **Garage**. You can store up to 16 characters for each name. Use the Programmer keys(**A-Z**) to type in the name of the light. Use the Select button to make spaces or delete characters.

CL24 LiteJet™ Programming Lights

STEP 5

LT 01 ▼:Next ▲:Prev
Dimmer? ◀Yes ▶:No

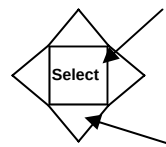


After you've entered the name, press either Select or ▼ to go to the next screen. Now you will begin to set the various attributes for this **Light**. You must first tell the CL24 if this **Light** should be a **Dimmer**. You can see what it is set for now by the blinking cursor on either "Yes" or "No". Make your selection using either ◀ or ▶. Then press Select or ▼ to go to the next screen.

NOTE: Do Not make fluorescents or motorized devices a DIMMER. This could damage those types of equipment.

STEP 6

LT 01 ▼:Next ▲:Prev
Dim Rate 5 Secs ◀ ▶

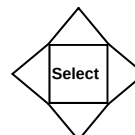


The Dim rate is the amount of time a Light will take to go from 0% to 100% while its button is being pressed and held. You shouldn't make this too short because you won't be able to finely adjust your levels. You also should not make it too long because you will get frustrated having to hold the button so long to change levels. 5 seconds is a good amount.

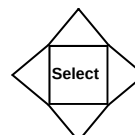
Press ◀ or ▶ to change the Dim Rate.

STEP 7

LT 01 ▼:Next ▲:Prev
Soft On? ◀Yes ▶:No



LT 01 ▼:Next ▲:Prev
Soft Off? ◀Yes ▶:No



The next 2 screens set the way a Dimming Light will function when it is toggled on or off with a button. If Soft On is set to **YES** then the Light will slowly fade up to its preset level when toggled on.

If **Soft Off** is set to **YES** then the Light will slowly fade to 0% when it is toggled off.

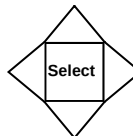
If **Soft On** is set to **NO** then the light will go immediately to its preset level when toggled on.

If **Soft Off** is set to **NO** then the light will go immediately off when toggled off.

CL24 LiteJet™ Programming Lights

STEP 8

LT 01 ▼:Next ▲:Prev
Soft Rate 5 Secs ◀ ▶

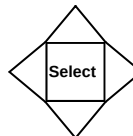


The Soft Rate is the amount of time a Light will take to go from 0% to 100% when it is toggled.

Press ◀ or ▶ to change the Soft On/Off Rate.

STEP 9

LT 01 ▼:Next ▲:Prev
Set Level 100% 0,1,..,9

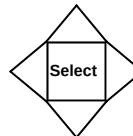


On this screen you can set a Preset Dim Level for the Light. This preset is the level the light will go to when toggled On. You can override this level to 100% by dimming the Light manually. This feature is for saving energy and extending the life of bulbs.

To change the level, just enter the percentage on the numeric keypad.

STEP 10

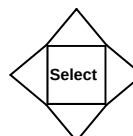
LT 01 ▼:Next ▲:Prev
ALL ON? ◀ ☒ Yes ▶:No



Set whether this Light will be part of the built-in All On scene.

STEP 11

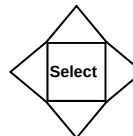
LT 01 ▼:Next ▲:Prev
ALL OFF? ◀ ☒ Yes ▶:No



Set whether this Light will be part of the built-in All Off scene.

STEP 12

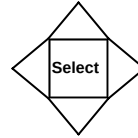
LT 01 ▼:Next ▲:Prev
Vacation? ◀ ☒ Yes ▶:No



Set whether this Light will be part of the Vacation Scene. When activated, the Vacation scene will cycle the selected lights on and off randomly to simulate a “lived in” look. When active, the Vacation Mode will only cycle lights between 5:00PM and 12:00AM Midnight.

STEP 13

LT 01 ▼:Next ▲:Prev
Alarm F.? ◀Yes ▶:No



Set whether this Light will be part of the Alarm Flash Scene.

Any Lights selected to be part of the Alarm Flash scene will blink on and off once a second as long as a contact closure input is maintained in Input #1 on the MCP.

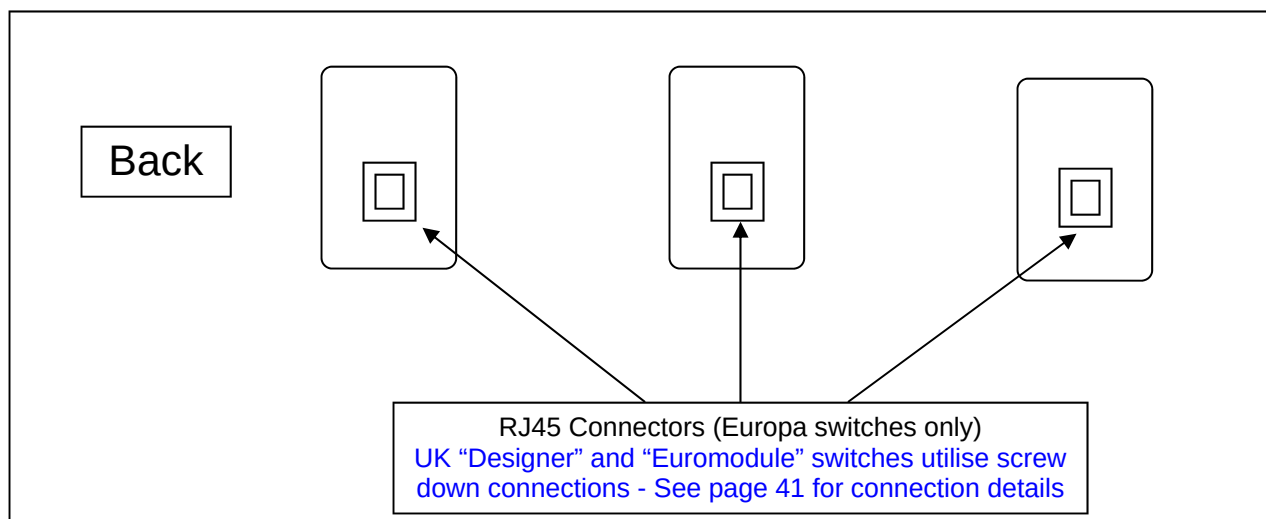
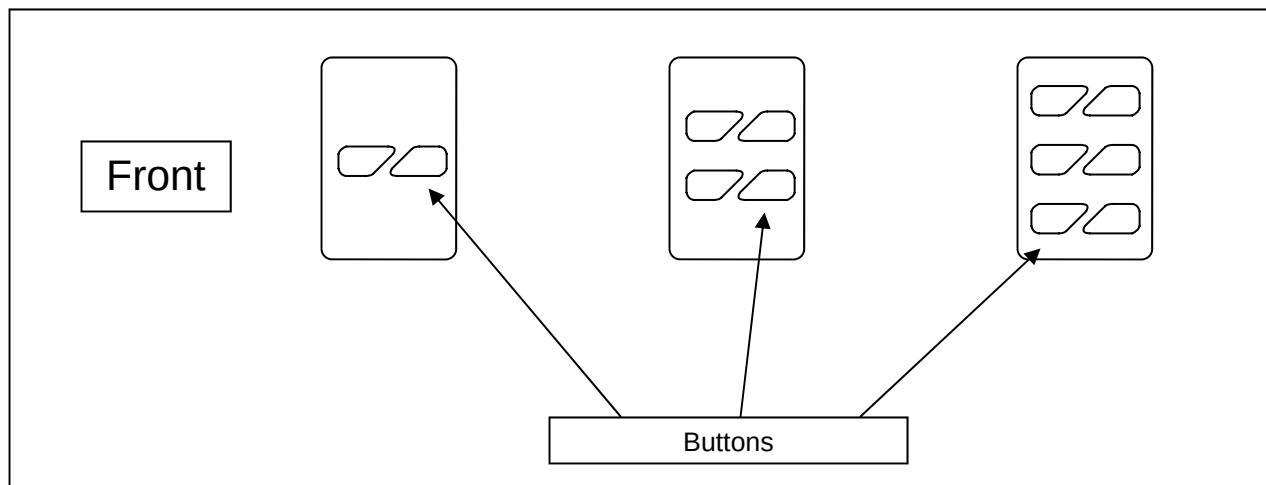
CL24 LiteJet™ Programming Buttons

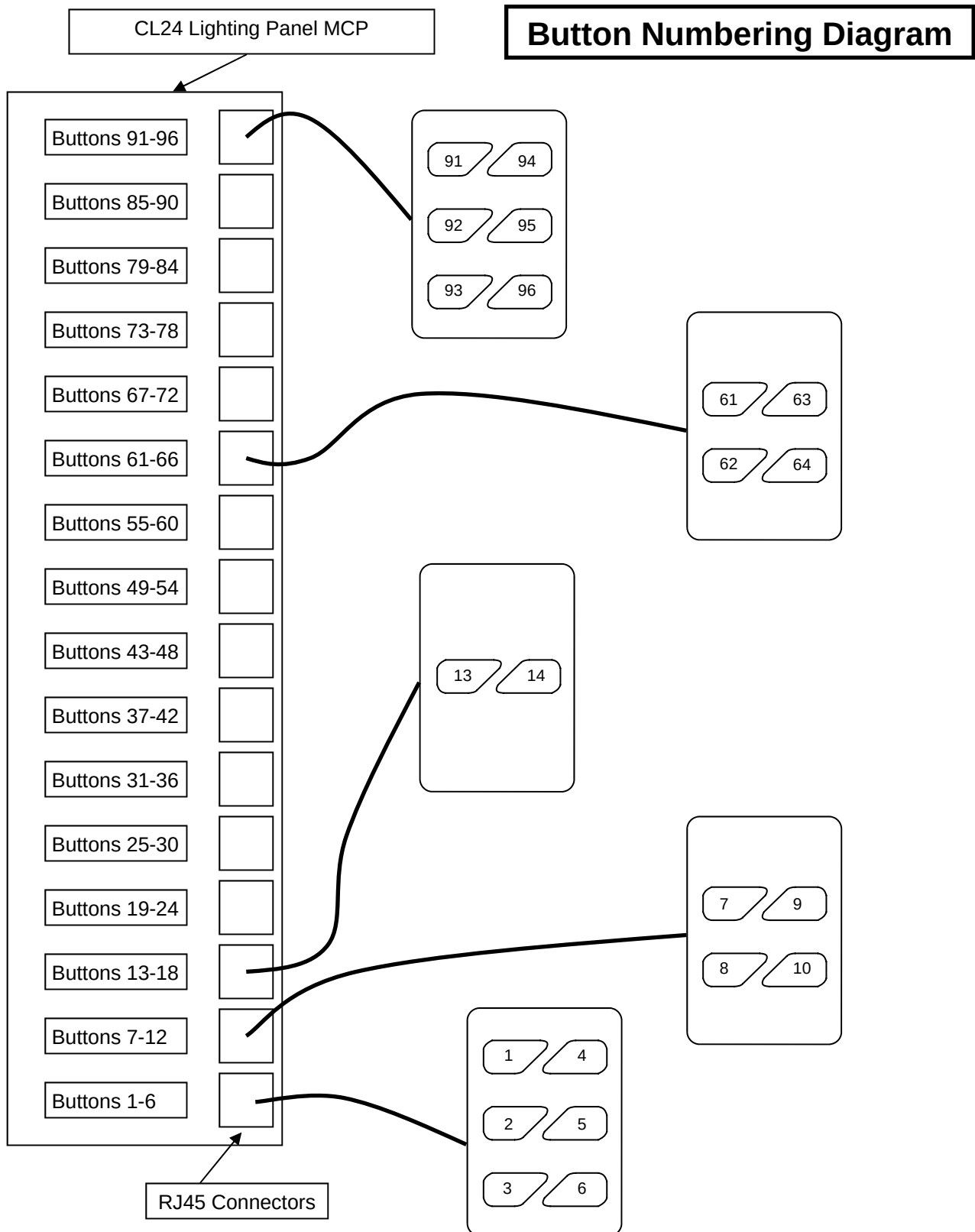
Buttons in the CL24 system are composed of 24 programmable buttons on the Display Panel and up to 96 buttons on the wall mounted keypads.

The CL24 MCP has 16 RJ45 keypad connectors. Each connector is capable of handling up to 6 buttons. If a 4 button keypad is plugged into one of these connectors, then 2 potential button addresses will not be used. Likewise, if a 2 button keypad is plugged in, then 4 potential button addresses will not be used.

Each RJ45 connector on the MCP has 6 fixed addresses for keypad buttons. The first connector is for buttons 1 through 6. The second connector is for buttons 7 through 12 and so on. The last connector is for buttons 91 through 96. (See diagram on page 12)

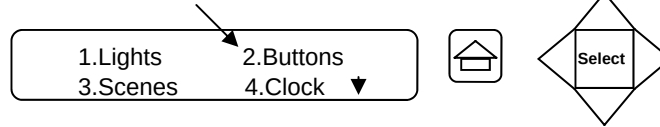
Each button can be assigned to an individual Load or Scene. The buttons can be easily removed and changed with different labeling. The CL24 kit contains 112 buttons with standard labeling. Custom labeled buttons can be ordered from CentraLite





CL24 LiteJet™ Programming Buttons

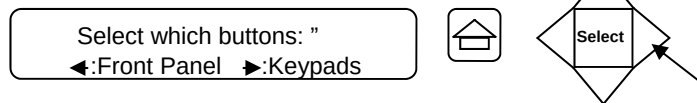
STEP 1



Begin at the initial programming screen.

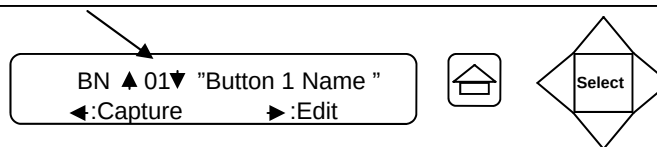
Press 2 to begin programming buttons.

STEP 2



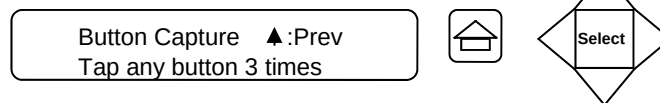
You can choose to program the Front Panel buttons or the buttons on the wall keypads. Let's start by programming the keypads. Press →.

STEP 3



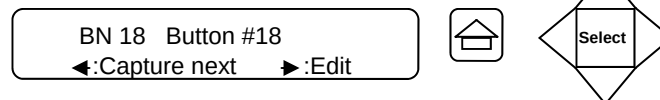
You can scroll through the buttons(1 thru 96) by pressing ▲ and ▼. If you don't know the address for the switch you want to program, then you can use the button Capture mode. Press ◀ to enter the Capture mode.

STEP 4



Now you can go to a keypad button and tap it at least 3 times. When you return to the display, you will see the address of the button you just pressed. You can now choose to Edit this button or capture a different one.

STEP 5

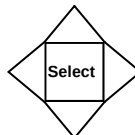


Now press → to Edit this button.

CL24 LiteJet™ Programming Buttons

STEP 6

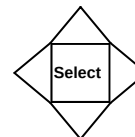
BN 5 SEL=Space ▼:Next
Name:"Kitchen"



After entering the name, Press ▼ to continue.

STEP 7

BN 5 ▼:Next ▲:Prev
◀ LIGHT ▶ :SCENE

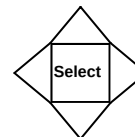


Buttons can either control a **Light** or **Scene**. Select which by pressing ◀ or ▶.

As soon as you select **Light** or **Scene** you are taken to the next screen. For this example we will choose **Light**.

STEP 8

BN 5 ▼:Next ▲:Prev
LT# ◀ 05 ▶ Chandelier

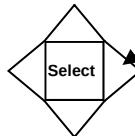


Now we must choose which **Light** to assign to this button. You can scroll through the Lights with ◀ and ▶. Now you can see why it is important that you name your Lights. The names of the Lights appear here to help you select the right one.

Once you have found the correct Light, press ▼ or **Select** to go to the next screen.

STEP 9

BN 5 ▼:Prev
◀:Standard ▶:Advanced



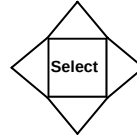
Now you can choose what settings to assign to this button. If you are not sure what to do here, just press ◀ for Standard. If you want to change the button settings press ▶ for Advanced.

For this example, we will choose **Advanced**.

CL24 LiteJet™ Programming Buttons

STEP 10

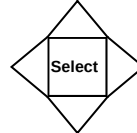
BN 5 ▼:Next ▲:Prev
Send Action? ◀:Yes ▶:No



Choose YES if you want the CL24 to send a notification out both RS232 ports whenever this button is pressed. Then press ▼ to continue.

STEP 11

BN Type ▼:Next ▲:Prev
1:Norm 2:Contact 3: Ramp



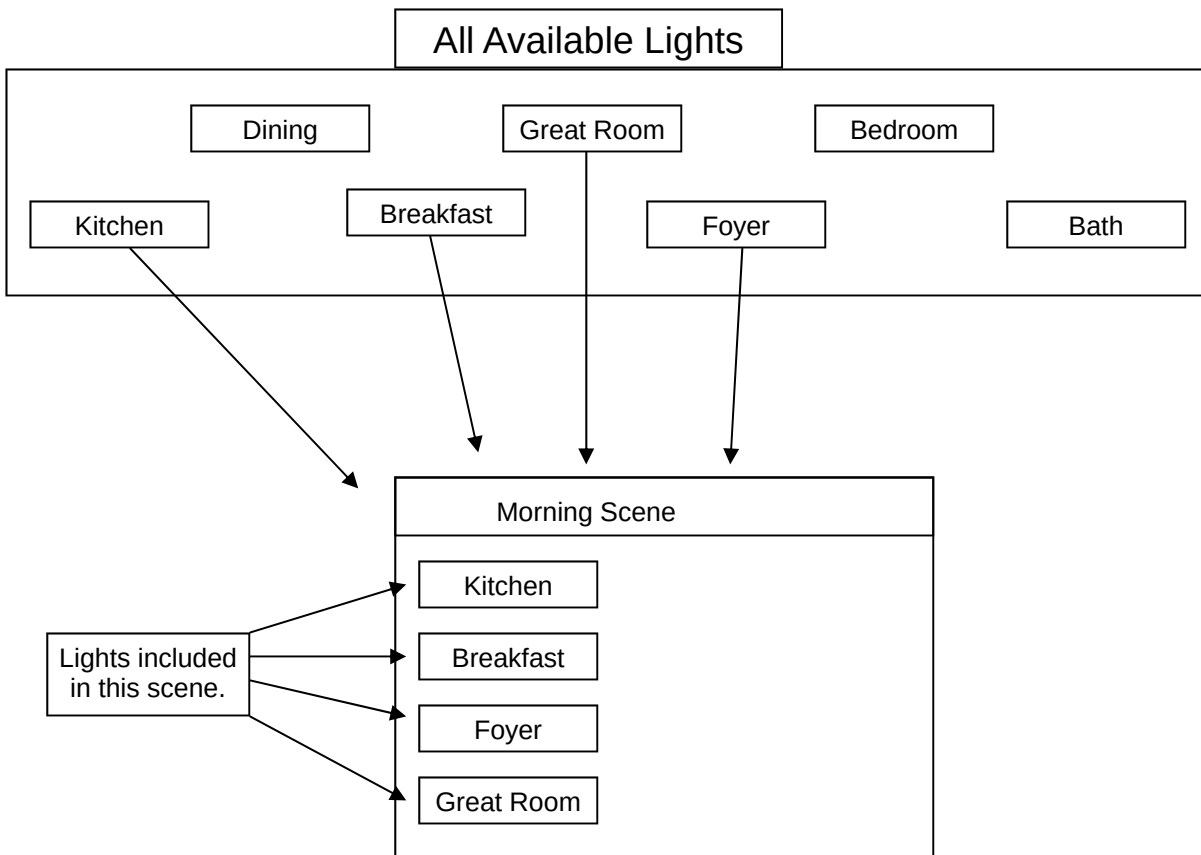
- 1: **Norm** = Normal Button. You can then choose Regular or Simple type. Regular recognizes long button presses for dimming and deactivating scenes. Simple will ignore any long button presses. Simple is useful for relay contacts which always give a long(> 1 sec) press. This mode is necessary for the remote control receiver option.
- 2: **Contact** = Contact button type. Use this mode when a door contact, or motion sensor is tied into this button. You can then select if the contact is normally open or closed.
- 3: **Ramp** = Raise/Lower. This type of button assigns a one way operating Mode (Up or Down) to a button. This is useful for raising or lowering drapes or blinds. It is also useful if you want raise and lower for a scene level.

CL24 LiteJet™ Programming Scenes

Scenes are groups of Lights with preset Dim Levels and Fade Rates. Scenes can be activated by buttons, clock events, or external commands via RS-232 ports.

EXAMPLE:

We will create a Morning Scene. The following represents all the Lights available in our example. First we need to choose which Loads will be part of the Scene. Any Lights which are not part of the Scene will not be affected when we activate or deactivate the Scene.



The first step in creating a Scene is complete. We have chosen which Lights are included in this Scene. Now we must configure how the Lights will behave when the Scene is activated. The settings for Lights include **Dim Level**, **Fade Rate**, and **Delay**.

Dim Level: What percentage of brightness will the Light go to when the Scene is activated.

Fade Rate: How long will the Light take to go from its current level to the programmed Dim Level. This rate is in seconds and represents how long the light would take to fade from 0% to 100%.

Delay: Each Load can be set to wait for a number of seconds before it activates. This is useful for Scenes like **Goodbye** where you may want a Garage light to stay on for a few minutes before turning off.

Morning Scene		
	Level	Fade Rate
Kitchen	75%	5 Secs
Breakfast	60%	9 Secs
Foyer	70%	3 Secs
Great Room	50%	16 Secs

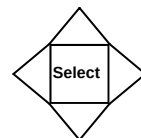
When this **Scene** is activated the loads will go to these levels at the specified **Fade Rates**.
When the **Scene** is deactivated, the loads will go to 0%.

Note that non-dimming lights can only be set to 0% or 100% and can not have a fade rate.

CL24 LiteJet™ Programming Scenes

STEP 1

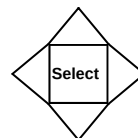
1.Lights 2.Buttons
3.Scenes 4.Clock ▼



Let's begin programming Scenes by selecting **3**.

STEP 2

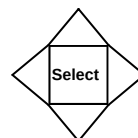
SN ▼05 ▲ "Power-up Scene"
◀:Capture/Test ▶:Edit



You can scroll through the scenes with the ▼ and ▲ keys. When you get to the scene you wish to work on press ▶ for Edit or ◀ to Capture Levels or Test the scene. For our example we will scroll to Scene 6 and press ▶ to Edit.

STEP 3

SN 06 SEL=Space ▼:Next
Name:"Welcome"

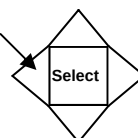


You will first type in the name of this scene using Display Panel keys and **SELECT** for spaces or to delete characters. For our example, we will make this scene "Welcome".

Press ▼ when finished entering name.

STEP 4

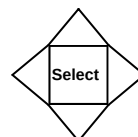
SN 06 "Welcome"
◀:Set Lights ▶:Advanced



We can now choose to setup the lights for this scene or go to some advanced settings. For our example, we will choose to start with setting up the lights.

STEP 5

SEL=Include ▼:Next ▲:Prev
LT ◀ 01 ▶ "GARAGE "Removed

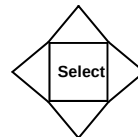


By default no lights are initially included in the scene. You must decide which lights will be included and which will be removed from the scene. You can scroll through the lights using ◀ and ▶. When you want to include the current Light, we press **SELECT**. The light is now included in the scene. If you hit **SELECT** again you will remove it.

CL24 LiteJet™ Programming Scenes

STEP 6

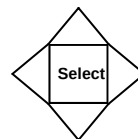
SEL=Remove ▼:Next ▲:Prev
LT ◀ 04 ▶ "KITCHEN "Included



A good method is to scroll through the lights, and quickly choose whether to include or remove them from the scene then scroll to the next light and so on. Once you've done this, you can go back through and setup each light individually. To setup what a light will do when the scene is activated, just scroll to that light and press ▼ or **SELECT**.

STEP 7

LT 04 ▼:Next ▲:Prev
Rate? 5 Sec ◀ ▶

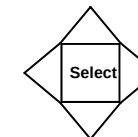


You will first choose what rate you want the light to transition. This rate is similar to the Dim and Soft rates for lights. It represents how long a light takes to transition from 0% to 100% when a scene is activated. Make this rate small if you want lights to quickly turn on or off. Make this rate longer if you want a smoother change for mood lighting. Press ◀ and ▶ to change the transition rate. You may need to change the rate several times and test the scene before you get the right one. Press ▼ to continue.

Note that you cannot change the rate of non-dimming lights.

STEP 8

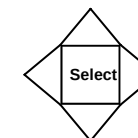
LT 04 ▼:Next ▲:Prev
Level? 50% 0,1,..,9



Next you will select what level you want this light to go to when this scene is activated. The "0,1,..,9" means that you enter this information in with the numeric keys on the display panel. Sometimes it's difficult to get just the right level by entering it here. We'll see shortly how we can "Capture" the levels for a scene. This means that we can manually set the levels in the rooms with the wall keypads and then get the CL24 to save these levels automatically. Press ▼ to continue.

STEP 9

LT 04 ▼:Next ▲:Prev
Delay 00000 Sec 0,1,..,9



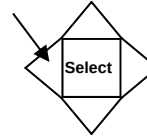
Next you can set a delay for this light. For example, if you enter 20 seconds, then **Light 04** will only activate 20 seconds after this **Scene** has been activated. This is useful for a Goodbye type scene among others. A Goodbye scene might turn off all the lights of the house. But you may want to wait 1 minute before turning off the Garage and Front Entrance lights to allow the homeowner time to get to the car. You would give these lights a delay of 60 seconds.

Repeat Steps 5 through 9 to program each light in a scene.

CL24 LiteJet™ Programming Scenes

STEP 10

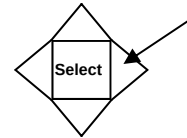
SN ▼ 06 ▲ "Welcome"
◀:Capture/Test ▶:Edit



When we've finished setting up lights, we will go to the initial screen for Scene 6 by pressing the key twice. Now we can Test the scene. Press ◀ to go to the Capture/Test screen.

STEP 11

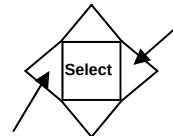
SN 06 "Welcome"
◀:Capture Levels ▶:Test



Now press ▶ to go to the Test screen.

STEP 12

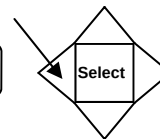
SN 06
◀:Activate ▶:Deactivate



You can now cycle Scene 06 On and Off by pressing ◀ and ▶. If the scene is not acting as expected, you can go back and try editing again. You can also go up one level to Capture Levels.

STEP 13

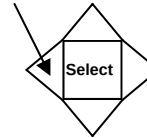
SN 06 "Welcome"
◀:Capture Levels ▶:Test



You can select ◀ to go to the Capture Levels Screen.

STEP 14

Are you sure?
◀:Yes Capture ▶:No

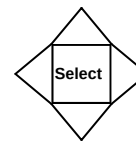


If you press ◀ the system will capture the current dim levels of the lights into your scene. It will overwrite any of the levels you have already programmed for this scene. This is why you are asked if you are sure.

Before pressing ◀ ,you can walk around the house and set the dim levels with wall keypads. When you are satisfied that you have set the levels correctly, come back to the programming panel and press ◀ . If you do not want to capture levels at this time, press ▶ .

STEP 15

SN 06 ▼:Next ▲:Prev
Levels Are Captured...



You have now captured dim levels for this scene. You can press ▼ to continue or ▲ to go back and capture again.

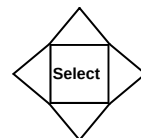
If you wish to assign a button to this scene, go to page 20.

CL24 LiteJet™ Programming Timed Scenes

There are 12 timed **Scenes** in the CL24. They are Scenes 30 through 41. The timed scenes activate any one of **Scenes** 1-29 at a particular time of day or based on sunrise and sunset.

STEP 1

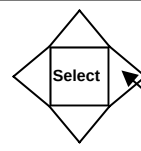
1.Lights 2.Buttons
3.Scenes 4.Clock ▼



To navigate to the times scenes, you first select "3.Scenes" from the main programming menu.

STEP 2

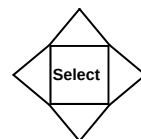
SN ▲ 30 ▼ "Timed Scene #01"
►:Edit



For our example, we will scroll to scene 30. This is the first timed Scene. Press ► to edit this timed scene.

STEP 3

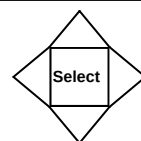
SN 30 SEL=Space ▼:next
Name: "Landscape On "



For our example, we will name this scene: "Landscape On". We will set this scene to turn on the Landscape Scene(9) at 7:30PM(1930). Press ▼ to continue.

STEP 4

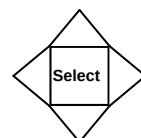
SN 30 ▼:prev
Enable Scene? ◀:Yes ▶:No



You must first Enable this scene. Timed events, by default, are not enabled. So press ◀ to enable this timed scene. The press ▼ to continue.

STEP 5

SN 30 ▼:next ▲:prev
SN ◀ 09 ▶ "Landscape "

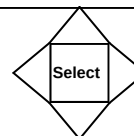


Next, you must choose which **Scene** will serve as the action for this **Timed Scene**. Scroll through the available **Scenes** using ◀ and ▶. When you've got your desired action **Scene**, just press ▼.

CL24 LiteJet™ Programming Timed Scenes

STEP 6

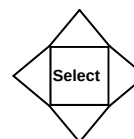
SN 30 ▼:next ▲:prev
Turn Scene? ◀N ▶:OFF



You must select whether to turn the action **Scene** On or OFF. Press ▼ to continue.

STEP 7

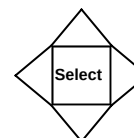
SN 30 ▼:next ▲:prev
Months of Year : ◀Jan▶SEL



The next step is selecting which months of the year this timed scene will function. By default, all months are selected. You can scroll through the months using ◀ and ▶. When the cursor is flashing on a month, that month is selected. To deselect the current month from this **Timed Scene**, press **SELECT**. The cursor will stop flashing over the current month. When you are finished, press ▼ to continue.

STEP 8

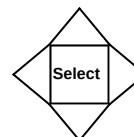
SN 30 ▼:next ▲:prev
Days of Week : ◀Sun▶SEL



The same logic applies to selecting the Days of the Week the **Timed Scene** will operate. By default, all days are selected.

STEP 9

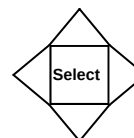
SN 30 ▼:next ▲:prev
◀Absolute ▶:Relative



The next choice is for how this **Timed Scene** will work: based on an **Absolute** time of day, or **Relative** to sunrise or sunset. For our example we will select **Absolute**.

STEP 10

SN 30 ▼:next ▲:prev
Hrs:Mins 19:35 0,1,..,9

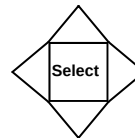


Now you set the time we want the **Timed Scene** to occur in 24 Hour format. In this case you want the **Timed Scene** to activate at 7:35PM. This is 19:35 in 24Hour format.

CL24 LiteJet™ Programming Clock

STEP 1

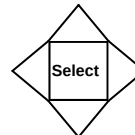
1.Lights 2.Buttons
3.Scenes 4.Clock ▼



To set up the real time clock, select “**4:Clock**” from the initial programming screen.

STEP 2

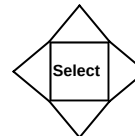
Date: ▼:Next
12/17/03 0,1,..,9



If the date is correct, you can go straight to the Day of Week screen by pressing ▼. Otherwise you can just type in the correct date. The Date is in Month/Day/Year format. Press ▼ to continue.

STEP 3

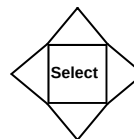
Day of Wk ▼:Next ▲:Prev
Su Mo Tu We Th Fr Sa ◀ ▶



You can choose the current weekday by scrolling with ◀ and ▶. The cursor will show which day of the week is currently selected. When you are finished, press ▼ to go to the Time screen.

STEP 4

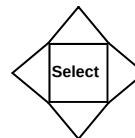
Time(24H) ▼:Next ▲:Prev
10:35:00 0,1,..,9



The time is set in 24 hour format: ex. 10:35:00 = 10:35 AM , 18:23:00 = 6:23 PM.

STEP 5

DL Savings ▼:Next ▲:Prev
◀:Enable ▶:Disable

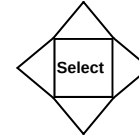


You will next choose whether Daylight Savings adjustments will be made automatically by the system. The default setting is **Enable**.

CL24 LiteJet™ Programming Clock

STEP 6

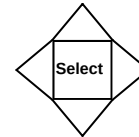
Latitude: ▼:Next ▲:Prev
30:40N 0,1,..,9,N,S



You must set your latitude and longitude for the sunrise and sunset functions to work properly. You can look up you latitude and longitude on any web search engine. Don't forget to enter **North** or **South**.

STEP 7

Longitude ▼:Next ▲:Prev
88:05W 0,1,..,9,W,E



Don't forget to enter **East** or **West**.

CL24 LiteJet™ Troubleshooting

Problem	Possible Solution
The CL24 system does not appear to be working at all.	Check the power to the system from the 12Vac 1000 MA transformer. Does the MCB supplying the transformer have power? When you have power you should see a GREEN LED on the MCP lit.
One of the keypads is not working.	Test your cable and RJ45 connections. Make sure it is programmed correctly. Do the LED's on the keypad button glow at all?
None of my keypads work.	Check power to the MCP. Make sure your connectors are terminated straight through.
The Touch Panel Programmer displays: "Turn OFF DIP SWITCH #1 on Master Control Panel"	You'll need to access the Dipswitches on the MCP. The MCP is at the bottom of the Relay Panel. Turn off the #1 switch.
On one of my relay modules, I have a Green LED lit but no Yellow LED.	Make sure the RED LED is lit. If not check the breaker. If the RED LED and the GREEN LED are lit but you have no YELLOW LED then you have a faulty relay. You'll need to replace it.
I have a Light that stays ON all the time no matter what I do.	If the Yellow LED stays lit even with the GREEN LED OFF, you have a relay that has failed closed. You'll need to replace it.
All Lights on the System stay ON no matter what I do.	Check the Manual Override switch at the bottom of the MCP. It may be in the Override (UP) position. If it is, then turn it to the Normal (DOWN) position.

CL24 LiteJet™ Lights Worksheet

Light #	Name	Dimmer?	Soft On?	Soft Off?	Dim Rate	Soft Rate	Include in All On?	Include in All Off?	Preset Level
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									
17									
18									
19									
20									
21									
22									
23									
24									

CL24 LiteJet™ Keypads 1-4 Worksheet

Keypad #	1	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	1				
Button	2				
Button	3				
Button	4				
Button	5				
Button	6				

Keypad #	2	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	7				
Button	8				
Button	9				
Button	10				
Button	11				
Button	12				

Keypad #	3	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	13				
Button	14				
Button	15				
Button	16				
Button	17				
Button	18				

Keypad #	4	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	19				
Button	20				
Button	21				
Button	22				
Button	23				
Button	24				

CL24 LiteJet™ Keypads 5-8 Worksheet

Keypad #	5	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	25				
Button	26				
Button	27				
Button	28				
Button	29				
Button	30				

Keypad #	6	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	31				
Button	32				
Button	33				
Button	34				
Button	35				
Button	36				

Keypad #	7	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	37				
Button	38				
Button	39				
Button	40				
Button	41				
Button	42				

Keypad #	8	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	43				
Button	44				
Button	45				
Button	46				
Button	47				
Button	48				

CL24 LiteJet™ Keypads 9-12 Worksheet

Keypad #	9	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	49				
Button	50				
Button	51				
Button	52				
Button	53				
Button	54				

Keypad #	10	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	55				
Button	56				
Button	57				
Button	58				
Button	59				
Button	60				

Keypad #	11	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	61				
Button	62				
Button	63				
Button	64				
Button	65				
Button	66				

Keypad #	12	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	67				
Button	68				
Button	69				
Button	70				
Button	71				
Button	72				

CL24 LiteJet™ Keypads 13-16 Worksheet

Keypad #	13	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	73				
Button	74				
Button	75				
Button	76				
Button	77				
Button	78				

Keypad #	14	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	79				
Button	80				
Button	81				
Button	82				
Button	83				
Button	84				

Keypad #	15	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	85				
Button	86				
Button	87				
Button	88				
Button	89				
Button	90				

Keypad #	16	Location:			
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	91				
Button	92				
Button	93				
Button	94				
Button	95				
Button	96				

Lights Worksheet Example

Light #	Name	Dimmer?	Soft On?	Soft Off?	Dim Rate	Soft Rate	Include in All On?	Include in All Off?	Preset Level
1	<i>Kitchen</i>	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>	5	2	<i>Yes</i>	<i>Yes</i>	80%
2	<i>Garage</i>	<i>No</i>	<i>No</i>	<i>No</i>	0	0	<i>Yes</i>	<i>Yes</i>	100%
3	<i>Dining</i>	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>	7	4	<i>Yes</i>	<i>Yes</i>	75%
4	<i>Bedroom</i>	<i>Yes</i>	<i>Yes</i>	<i>No</i>	6	3	<i>No</i>	<i>Yes</i>	80%

Keypads Worksheet Example

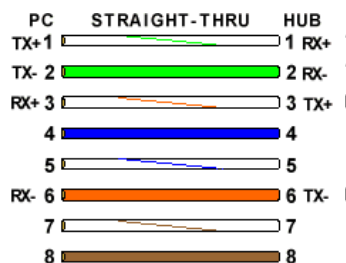
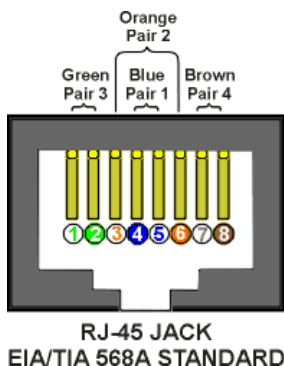
Keypad #	14	Location:	<i>Foyer</i>		
		Button Name	Light or Scene?	Light or Scene #	Notes
Button	79	<i>Chandelier</i>	<i>Light</i>	5	<i>Foyer Chandelier</i>
Button	80	<i>Front Porch</i>	<i>Light</i>	21	<i>Front Porch Light</i>
Button	81	<i>Landscape</i>	<i>Scene</i>	11	<i>Landscape Scene</i>
Button	82	<i>Welcome</i>	<i>Scene</i>	6	<i>Welcome Home Scene</i>
Button	83				<i>NOT USED</i>
Button	84				<i>NOT USED</i>

Button -	Either on the Touch Pad Programmer (TPP) or Wall Keypad. Can be programmed to control a single Light or Scene (group of lights).
Dimmer -	A control such as the CL24 solid-state relay which can vary the intensity of a lighting load.
Keypad -	A wall mounted control with either 2, 4, or 6 Buttons. The Keypad is connected to the Master Control Panel (MCP) with CAT5 cable.
Master Control Panel -	The “brain” of the CL24 system. The MCP takes in control inputs from buttons and sends control outputs to solid-state relays, fans, or low voltage relays.
Relay -	The CL24 uses solid-state relays which are capable of dimming incandescent, magnetic low voltage, and some electronic low voltage lighting fixtures. CL24 relays can turn on and off any electrical device (as long as the power draw is within the relay rating).
TPP -	The CL24 LiteJet Touch Pad Programmer is used to configure the CL24 LiteJet system as well as control 24 functions with the buttons on the front panel.
RS232 -	A communication method used for programming the CL24 LiteJet from a computer or interfacing CL24 LiteJet with other control systems.
RS485 -	Another communicating method used for controlling other devices or receiving commands from other control systems. One of the CL24s communication ports can be configured for RS485.

UK “Designer” & “Euromodule” keypads

The NEW UK style “Designer” and “Euromodule” switches and keypads do NOT use RJ45 connectors, they have screw down connections as shown below.

For the **LiteJet** system, wire the RJ45 connectors according to the TIA 568A standard shown below:
Stranded CAT5e patch cable is strongly recommended to avoid conductor breakage.



Wire colours

White/Green stripe
 Green
 White/Orange stripe
 Blue
 White/Blue stripe
 Orange
 White/Brown stripe
 Brown

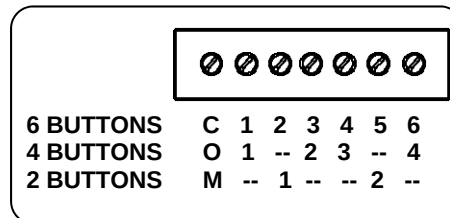
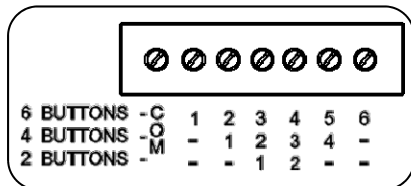
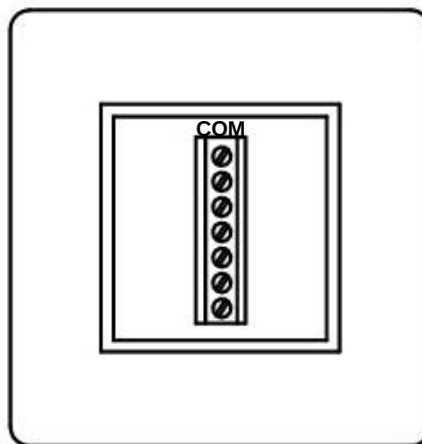
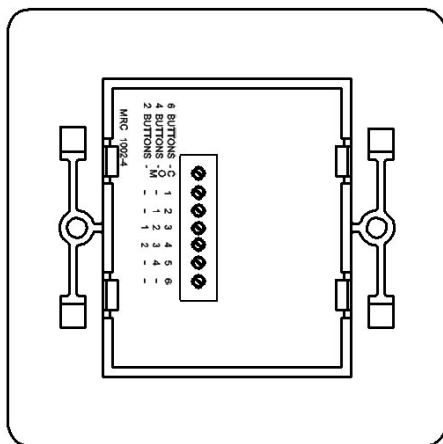
Connections

COMMON
 Button 1
 Button 2
 Button 3
 Button 4
 Button 5
 Button 6
 COMMON

UK “DESIGNER” connections:

“EUROMODULE” connections

NOTE! - The connections are NOT the same for the 4 button and 2 button switches



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CL24 LiteJet™

UK Installation and Programming Guide